

PEDAGOGY OF COMMERCE AND ACCOUNTANCY - I

TAMILNADU TEACHERS EDUCATION UNIVERSITY

Comprehensive and exhaustive coverage of the latest semester
(UNDER CBCS SEMESTER PATTERN 2021-2022)

B.Ed. (1st Year)
1st Semester

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COURSE OBJECTIVES

- CO1. Understand the Aims and Objectives of teaching Commerce and Accountancy.
- CO2. Comprehend instructional objectives for a lesson.
- CO3. Gain mastery of the teaching skills.
- CO4. Identify various methods in teaching Commerce and Accountancy.
- CO5. List various resources in teaching Commerce and Accountancy.

UNIT-I AIMS AND OBJECTIVES OF TEACHING COMMERCE AND ACCOUNTANCY

Commerce: Meaning, Nature and Scope – Aims and objectives of teaching Commerce in schools: Instructional objectives and Behavioural Objectives – Need and Importance of Instructional Objectives. Bloom's Taxonomy of Instructional Objectives: Cognitive, Affective and Psychomotor Domains, Revised Bloom's Taxonomy 2001 (Anderson & Krathwohl) Interrelation among the domains – Correlation between subjects – Values of Teaching Commerce.

UNIT-II: TEACHING SKILLS

Micro-Teaching: Concept, Definition, Steps, Cycle - Micro-teaching Vs Macro-teaching - Skill of Set Induction - Skill of Explaining, Skill of Questioning, Probing skills, Skill of Stimulus Variation, Skill of Reinforcement, Skill of non-verbal clues, Skill of Closure - Link lesson – Model episode.

UNIT - III: APPROACHES OF TEACHING COMMERCE AND ACCOUNTANCY

Approaches of Lesson Planning - Steps - Organizing Teaching: Memory Level (Herbartian Model), Understanding Level (Morrison teaching Model), Reflective Level (Bigge and Hunt Teaching Model) – Unit Plan – Lesson Plan Writing.

UNIT-IV: METHODS OF TEACHING

Teacher-centred Instruction: Lecture method, Demonstration and Team teaching – Learner-centred Instruction: Self-learning, Forms of Self-Learning: Programmed Instruction, Computer Assisted Instruction, Keller Plan, Project Method, Activity Based Learning (ABL), Active Learning Method (ALM)-Mind Map and Advanced Active Learning Method (AALM).

UNIT-V: INSTRUCTIONAL MEDIA

Classification of Instructional Media in Commerce and Accountancy – Use of Mass media in classroom Instruction. New Emerging Media: Tele-Conferencing, Learning, Flipped Classroom, Artificial Intelligence and Augmented Reality - Community Resources: Fieldtrips - Commerce Exhibition/Fair - Commerce Resource Centre - Commerce Club - Qualities of a good commerce textbooks - Qualities of a good Commerce teacher.

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UNIT 1

AIMS AND OBJECTIVES OF TEACHING COMMERCE AND ACCOUNTANCY

1.1 COMMERCE: ITS MEANING AND DEFINITION

It is not very long, since commerce has developed to a unique discipline with a firm standing of its own. Instead of being considered as one of the various dimensions of Economics, it has now emerged as a subject of study having its own features resulting in enormous scope and applications. It has also a variety of social functions to serve. The commerce teacher can transmit the spirit and trends of this 'science' with a social orientation and can develop the related understandings, attitudes, values, skills, etc. among the students only if he internalizes all these aspects by himself. As such, an attempt is made in the discussion that follows, to clarify the meaning, nature and characteristics of commerce, as well as its scope and functions emerging out of these characteristics.

1.1.1 Meaning of the term Commerce

The term 'commerce' as it is understood in modern times, conveys a specific meaning. The meaning of the term 'commerce' represents that particular domain of knowledge which deals with the concepts, principles, theories, processes and skill that could be

applied in the proper conduct of all tasks and transactions related to commercial activities. This is followed by the planning for convenient but cheap means of storing the goods, adopting scientific procedures for their trade and making provisions for effective distribution and customer services. It would also involve issues of financial provision, financial stability, insurance against various hazards, etc.

1.1.2 Definition of the Term Commerce

According to James Stephenson, "commerce is concerned with exchange of goods, with all that is involved with the buying and selling of goods, at any stage in their progress from raw materials to finished goods in the consumer's hands. It covers not only functions of buying, selling and handling of goods, but also the many services which must be provided to finance, insurance, store and transfer of goods in the course of this exchange."

According to Dr. Evelyn Thomas, "commerce is a term that embraces all those functions involved in the making, buying, selling and transport of goods."

1.2 COMMERCE EDUCATION: MEANING, DEFINITION AND IMPORTANCE

Any teacher who is expected to transact the curriculum of a subject to his students has first to develop a thorough understanding and meaningful insights regarding all aspects concerning that subject. A subject is accepted as a unique discipline for study because it conveys specific meanings and also possesses important characteristics which in turn lead to worthwhile functions to be served for the benefit of the society. The subject can be dealt with, in such a way as to fulfill the demands posed by the nature, scope and importance of the discipline, only if the teacher understands and appreciates these. This is very true of Commerce also.

1.2.1 Meaning of Commerce Education

Commerce education is directly concerned with the day to day life of the students. Even then it is necessary to define commerce education. According to Herrick, "commerce education is that form of instruction that both directly and indirectly prepares the businessman for his calling". In Herrick's view commerce education is preparation of a businessman. It includes all types of education which makes one person to become a great businessman

1.2.2 Definition of Commerce Education

In 1922 Lyon stated commerce education as "Any education which a businessman has and which makes him a better businessman is for him a commerce, no matter whether it was obtained in the walls of a school or not". According to Lyon curriculum for commerce education is all the activities of businessman and the ways by which he became a great businessman. To him that education is not restricted to the schools only since the student can even get his education outside the school i.e., from the society.

In 1933 Nichols defined commerce education as follows, 'commerce education is a type of training which while playing a part in the achievement of the general aims of education on any given level, has for its primary objective the preparation of people to enter upon a business career or having entered upon such a career to render more efficient services there in and to advance from their present levels of employment to higher levels'.

In 1950 Shields defined that real business education is economic education, Economic education not of the academic sort long on theory and short on facts, but economic education which will give the students a knowledge of the basic realities of business life and relationship. The basic realities of business of economic and without a thorough grounding awareness of economic problems much of the material included in secondary school business courses is purely additive and essentially superficial....We cannot place technique and socio-business subjects on a dual headed definition of the field but must recognize that certain elements must be most emphasized and these I take to be the economic factors". According to Shields commerce-education is based upon the economic education. The main frame work of the syllabi in commerce education is provided by economics.

1.2.3 Importance of Commerce Education

The main purpose of commerce education is to provide knowledge about the commerce and to prepare the student for vocational competency including training and development of skill. Commerce education is useful for the students to understand the various aspects of changing of the ownership of goods and to know about the various aids which facilitate the central process of exchange. Commerce education is aimed at giving adequate

knowledge about the wholesale trade, retail, export trade, import trade and entre-port trade. More-over it provides some knowledge about the movement of goods etc., Transport, Communication Insurance, Ware-housing, Money, Banking & Finance and Mercantile Agencies.

The main purpose of teaching Accountancy is to make the students understand the importance of book keeping its need and accuracy, neatness, orderliness, responsibility, honesty etc. This helps the students to know how to prepare and interpret simple financial statements and reports. It aids the students can understanding the posting of business transactions in the various forms of Accounts books such as journal, ledger, and other subsidiaries books etc. The commerce education helps the students to draw conclusion about the financial position of the organizations.

Many technical, industrial, and scientific developments are taking place. As these developments goes on the demand for efficient office workers will increase rapidly. More and more employment opportunities will be available at the executive and supervisory level as well as at the lower level. Commerce education provides proper training for executives and supervisory and lower level workers. The college programmes of commerce education will provide education for lower level such as typist, stenographers, book-keepers, file clerks, cashier etc. Division of labour and specialization leads every country not to be a self-sufficient one. Therefore many commercial transactions are carried out between one country and another. So the Foreign trade becomes inevitable, under such circumstances the importance of Foreign trade for the development of every country is imminent. In order to carry on the foreign trade effectively commerce education is an essential one. Commerce education provides knowledge about foreign trade, foreign exchange etc. It provides efficient supervisory personnel and the other able hands at lower level to assist the top executives. It also plays a vital role in the modern economic development. Thus it is very important at the higher secondary level.

Fulfilling the present day needs

Commerce is a dynamic process that brings together technology, natural resources and human initiatives in a constantly changing global environment. Hence, commerce education has

undergone a tremendous change in the wake of Liberalization, privatization and Globalization. Computerized systems are fast replacing other systems.

E-commerce and other related concepts, therefore, have been introduced as part of the higher secondary curriculum. It helps students to appreciate that commerce is an integral component of society and also to develop an understanding of many social and ethical issues.

With the fast changing economic scenario and business environment the commercial world is in a state of continuous flux. In tune with circumstance, commerce education has carved out a place for itself at the higher secondary and higher education stage. Its curriculum content should give students firm foundations in basic accounting principles and business studies to keep them abreast with the changes in its methodology concerning particular aspects of the subject. This would help to develop among students traits such as logical reasoning, careful analysis.

1.3 NATURE OF COMMERCE

Whether Commerce is a Science or Art: In fact, commerce is both science and art.

1. Commerce as an Art.

The terms 'Art' means the application of knowledge for achieving the solutions of some practical problems.

For J. N. Keynes "An art is a system of rules for attainment of a given end."

2. Commerce as a Science:

The term 'Science' refers the following characteristics:

1. Systematic study of knowledge which traces the relationship between cause and effect.
2. Systematic collection, classification and analysis of data/facts.
3. Its phenomena must be capable of being measured accurately in terms of money.

Like Science, theories and principles of commerce are universal and can trace the relationship between cause and effective easily. Commerce has a unique power of forecasting. Thus, we can say that commerce is also like science.

Commerce also suggests the practical application of knowledge for achieving definite ends. For instance, a firm suffers from high cost of its production; its result is the decline in the

demand of the goods in the market. With a view to achieve the objective of low cost of goods/production, a firm owner may take the help of planning; in fact, the path of planning to reduce the cost of production is an art because it refers the practical application of knowledge with the view to achieve the lowest cost of the production/goods.

Thus, we can say that commerce is both a science and an art because as a science, it formulates certain rules, principles and theories while as an art, it gives them a practical shape.

1. Commerce Education provides the occupational training to the students like clerical, accounting, secretarial and
2. It develops the human values in the students like patience, honesty, co-operate and brother etc. These human sale etc. qualities will help a business man to prosper his business rapidly.
3. Commerce Education' also helps in developing international understanding, because business person come in contacts with foreign businessmen and learn a lot of other's culture and manners too, while doing the business with foreign businessmen.
4. Commerce students receive the practical knowledge of commerce that helps in the future life of commerce students when they begin their own business.
5. Commerce education' at + 2 level will help a student in being a good consumer in future even though he does not continue his education after + 2.
6. Through commerce education, a businessman can also easily know about what is going on in the foreign markets. He can bring new technology in his farm or company, because he has up to date knowledge of current economic and businessman events of internal and external trades.
7. At+ level, different skills in commerce students can be developed as short hand, typing and practical knowledge of accountancy etc.
8. Commerce education teaches the students how to accumulate the money and how to invest that in the business.
9. Commerce education also provides the knowledge of diverse vocation as banking, office management, auditing, post office and insurance etc.
10. Commerce education prepares a number of good consumers for the producer.

1.4 SCOPE OF COMMERCE

Commerce education has a very wide scope and it covers almost all the aspects of human and national life.

According to Khan-"Commerce education got some impetus in conjunction with growth in the volume of business. The constantly growing demand for typists, stenographers, book keepers and clerical works commerce curriculum a very desirable addition to secondary school programme. The tremendous spurt in commerce and industry and planned economic development of the country made the possession of commercial competencies. Essential to commercial carrier, now, there is a great demand for office workers who can type reality and accurately, take and transcribe notes quickly and correctly keep financial records and operate business machines effectively. In addition it has been discovered that typewriting and other commercial subjects can make valuable contributions to the general education or the students.

"Keeping this view, the scope of commerce education can be mentioned like wise.

Scope of Commerce

Under the scope of commerce, we study:

1. The subject-matter of commerce.
2. Whether commerce is a science or art or both.

The Subject-matter of Commerce

The subject-matter of commerce-we can divide it into two Categories for our convenience:

1. Trade,
2. Auxiliaries to Trade.

Trade

An activity of buying and selling or exchanging goods or services between peoples or countries in monetary terms.

1. Home-Trade/Inland Trade
2. Foreign Trade/External Trade

1. **Home-Trade:** Home trade is also called as internal trade and refers to the buying and selling of goods within the boundaries of a nation. The payments for purchased goods are made in the national currency, i.e. in rupees (India). The personal that generally serve to establish links between producer and consumer, are:

- (i) Wholesalers,

- (ii) Retailers,
(iii) Mercantile Agents.

Home Trades Classification

Home trade may be classified into two categories:

- (i) Wholesale Trade,
(ii) Retail trade.

(i) **Wholesale Trade:** It connects purchased and sale of goods in huge amounts and low prices to shops and business rather than the selling of goods in shops to customers. The wholesale trade is a link between the producer and the retailer.

(ii) **Retail Trade:** It indicates the selling of goods to the public, usually in small quantities. The retail trade activities are generally performed through:

Shops	-	Super Bazar
Vendors	-	Self-service Stores
Multiple Shops	-	Peddlers
Cooperative Stores	-	Mail Order Houses.
Stall Holders	-	Departmental Stores

Subject matter of Commerce at +1 and +2 Stages

Dr. M. S. Khan: Says that "The adoption of 10+2 pattern of education has had a significant influence on public commerce education. Earlier, commerce subjects were taught in secondary schools from IX.... But under the new pattern of education known as 10+2, commerce subjects will be taught from class XI. It may be noted that in the examinations scheme of certain boards of education, commerce has been incorporated at the 10th class stage as an elective subject or this may be taken by the students as an extra subject. However, the courses of commerce at the XI and XII classes' stage is highly advanced."

The Central Board of Secondary Education, Delhi (2002-03) has revised the syllabus and outlined the following subject matter in Business Studies and Accountancy.

Subject-matter of Commerce Business Studies Foundations of Business:

1. Nature and Objective of Business
2. Social Responsibility and Business Entries
3. Structure of Business
4. E-commerce and Outsourcing of Services
5. Service Sector and Business
6. Banking and Insurance

7. Communications and Warehousing
8. Sole Proprietorship and Joint Hindu Family
9. Partnership and Cooperative Society
10. Company
11. Choice of Form of Business Enterprise and Starting a Business
12. Sectoral Organization of Business (Public Sector Enterprises)
13. Global Enterprises

Corporate Organization-Finance and Trade

1. Formation of a Company
2. Source of Business Finance
3. Institutional Finance
4. Internal Trade
5. Types of Retail Organizations
6. External Trade
7. Export Import Procedures
8. Project Report

Business Studies

Part A: Principles and Functions of Management:

1. Nature and Significance of Management
2. Principles of Management
3. Business Environment
4. Planning
5. Organizing
6. Staffing
7. Directing
8. Controlling

Part B: Business Finance and Marketing:

1. Financial Management
2. Capital Market
3. Marketing Management
4. Consumer Protection

1.5 NEED AND IMPORTANCE OF COMMERCE AND ACCOUNTANCY EDUCATION

The main purpose of commerce education is to provide knowledge about the commerce and to prepare the student for vocational competency including training and development of skill. Commerce education is useful for the students to understand the various aspects of changing of the ownership of goods and to know about the various aids which facilitate the central process of exchange. Commerce education is aimed at giving adequate knowledge about the wholesale trade, retail, export trade, import

trade and entre-port trade. More-over it provides some knowledge about the movement of goods etc., Transport, Communication Insurance, Ware-housing, Money, Banking & Finance and Mercantile Agencies.

The main purpose of teaching Accountancy is to make the students understand the importance of book keeping its need and accuracy, neatness, orderliness, responsibility, honesty etc. This helps the students to know how to prepare and interpret simple financial statements and reports. It aids the students can understanding the posting of business transactions in the various forms of Accounts books such as journal, ledger, and other subsidiaries books etc. The commerce education helps the students to draw conclusion about the financial position of the organizations.

Many technical, industrial, and scientific developments are taking place. As these developments goes on the demand for efficient office workers will increase rapidly. More and more employment opportunities will be available at the executive and supervisory level as well as at the lower level Commerce education provides proper training for executives and supervisory and lower level workers. The college programmes of commerce education will provide education for lower level such as typist, stenographers, book-keepers, file clerks, cashier etc. Division of labour and specialization leads every country not to be a self-sufficient one. Therefore many commercial transactions are carried out between one country and another. So the Foreign trade becomes inevitable, under such circumstances the importance of Foreign trade for the development of every country is imminent. In order to carry on the foreign trade effectively commerce education is an essential one. Commerce education provides knowledge about foreign trade, foreign exchange etc. It provides efficient supervisory personnel and the other able hands at lower level to assist the top executives. It also plays a vital role in the modern economic development. Thus it is very important at the higher secondary level.

1.6 AIMS AND OBJECTIVES OF TEACHING COMMERCE IN SCHOOLS:

Education is a process of bringing about changes in an individual in a desired direction. It is a process of helping a child to develop his potentialities to the maximum and to bring out the best from within the child. To bring about these changes we teach

them various subjects at different levels for school. Science as a subject is included in the school curriculum from the very beginning. Before thinking any decision about teaching science? What are the goals and objectives of teaching science? What changes does science teaching bring about in the behavior of the students?

1.6.1 Aims, Goals and Objectives - Meaning

A goal is the ultimate target in life which an individual tries to achieve. Goals are basically long term objectives which require a long period of time to be achieved. For example, acquiring knowledge about some specific things within a pre specified time limit would come under the purview of objectives.

The goals are the final destination expected by the pupils after learning different subjects in the school curriculum. The purposes of learning different subjects are called aims. A number of subjects are taught to achieve the aims and to reach the goals.

Students are potential human resources required to be enveloped into educated, sound, skilled and efficient citizens who will fulfil the aims of social, economic, political and technological development of the society

The aim is the term used in wider sense and broader concept. For Eg: one of the aims of education to eradicate illiteracy. This destination cannot be reached in one year or in short period. It takes many years sometimes aims cannot be achieved within the school years; but the school may give direction or may help to achieve the aims.

The aim of education is to provide opportunities for personal all-round development of individuals, to equip them with up to date and sound knowledge of science and technology and make them capable of using science and technology for the betterment of human society.

Aims are common for all subjects. Aims provide directions in education. Aims provide guidance to the entire educational system so aims of education can be stated but cannot be achieved within the school programme in short duration.

How do we achieve aims? It is by framing objectives, objectives are the possible points of achievements to reach the goal. It is possible to state the expected achievements in every class and in every subject. Objectives are different for different subjects. By realizing objectives through different subjects it is

possible to achieve aim in the long run. So objectives are the short term processes framed to reach the long term product, the aim.

Teaching will be effective only if the teachers have the knowledge of aims and objectives of teaching their subjects.

1.6.2 Difference between Aims & Objectives

S.No	Aims	Objectives
1.	Aims are long term statements of purpose that may be achieved over a long period of time, say one or more years. <i>Example</i> - Developing a sense of inquiry in learners is an aim that may be achieved over a number of years.	Objectives are bound in a short and specified time, say one teaching learning period or during teaching learning of one chapter. <i>Example</i> - conducting an activity to generate a question, why do the <i>Electric charges produced by rubbing two rubber balloons are static</i> , can be one step towards achieving this aim. It is achievable in short time and gives detailed picture of the step you would take.
2.	Aim is a foreseen end.	Objectives are influenced by aims.
3.	Aims are broader in sense.	Aims are narrower in sense.
4.	It may or may not be easily observable and measurable. For example, <i>developing interest/curiosity of the learner in science</i> .	Learning objectives are easily observable and measurable. For example <i>applying Ohm's law to calculate resistance of a conductor</i> .
5.	Assessment of all the aims may not be objective in nature.	Assessment of all the aims may be objective in nature.

1.6.3 Goals/Aims for Physical Science Teaching

Considering the individual teacher and his/her own reason for teaching science it would become impossible for any educational system to provide facilities for each teacher to achieve his/her goals. Therefore, goals are decided at the national level.

Various commissions and committees on education set up by Government of India after Independence emphasized the teaching of science from the primary level.

Following are the aims of teaching physical science:

- Knowledge
- Understanding
- Application
- Skills
- Abilities
- Attitudes
- Interest
- Appreciation
- Effective Use of Leisure time
- Training for better living
- Basis for vocation

1. Knowledge:

This is one of the major aims of teaching biology. Pupils studying biology should acquire the

- ❖ Knowledge of facts, events, persons, places etc.
- ❖ Knowledge of scientific terminology.
- ❖ Knowledge of ways and means of dealing with specifics.
- ❖ Knowledge of conversations.
- ❖ Knowledge of classification.
- ❖ Knowledge of trends and sequences.
- ❖ Knowledge of natural phenomena.
- ❖ Knowledge of theories.
- ❖ Knowledge of rules.

2. Understanding:

The teaching of biology should develop the understanding of biological facts, principles, processes etc. pupil should develop the understanding of the functional aspects in biology.

3. Application:

Pupil should apply the knowledge and understanding in new situations. In our day to day life the knowledge of microscope, nutrition etc., may be applied for healthy living.

4. Skills:

Psychomotor development it given importance through biology. Pupils develop the following skills:

- ❖ General skills like reading and writing.
- ❖ Communication skills like speaking and listening.
- ❖ Social skills like working in group, co-operation, service mind, emotional stability, sense of belongingness etc.
- ❖ Library skills like searching for reference.
- ❖ Laboratory skills like setting up apparatus, observation, developing, and preservation skill, handling, apparatus.
- ❖ Mathematical skills like ranking, dealing with symbols.
- ❖ Aesthetic skills like preparing charts, models, diagrams, etc.
- ❖ Abstract skills like recognition and classification of theories.

5. Abilities:

Pupils should develop the abilities of interpreting, analyzing, generalizing and predicating the biological events, the abilities to organize exhibitions clubs, quiz programmes etc. Abilities to locate and collect information regarding biological facts phenomena etc.

6. Attitudes:

Teaching of biology should aim at developing scientific attitude. A man with scientific attitude has the following characteristics.

- ❖ Open mindedness.
- ❖ Respecting other's point of view.
- ❖ Curiosity.
- ❖ Objectivity.
- ❖ No faith on superstitions.
- ❖ Suspended judgment.
- ❖ Belief in cause and effect relationship.
- ❖ Honesty.
- ❖ Impartiality.

7. Interests:

The teaching of biology should aim at developing some interests in reading scientific literature, hobbies, clubs, nature etc. Their needs the rationalization of desires and sense of fun among children. So biology should channelize them through the following activities.

- ❖ Collection of biological specimen like seeds, flowers, leaves, shells etc.

- ❖ Preparation of oil, soap, nail polish, greeting cards etc.
- ❖ Making improvised apparatus
- ❖ Visiting places of scientific value.
- ❖ Involving in science club activities.
- ❖ Reading scientific books, journals, periodicals etc.
- ❖ Writing articles of scientific value.
- ❖ Enjoying activities of science.
- ❖ Preparing models

These activities may be provided by the teacher to foster interest in biology. Teachers may organize science fairs, exhibitions, excursion, museum, and for library reading etc.

8. Appreciations:

After studying biology, pupils should develop admiration of biology in the following areas.

- ❖ History of biology
- ❖ Biographers of biologists
- ❖ Impact of biology on our living
- ❖ Contributions of biology in the progress of humanity
- ❖ Values of scientific method
- ❖ Natural phenomena

9. Useful spending of leisure time:

If pupils are not guided for the proper spending of leisure time they may become idle, they may become the cause of indiscipline. Teaching of physical science enables the pupils to preparation of materials namely kunkum, eyetex, soap, ink, soap nut powder, castor oil, juice preservation etc., may also fetch money. These hobbies may also lay foundation for their vocation.

10. Training for better living:

Physical Science gives the pupil knowledge about the taking care sanitation, pollution etc., and this knowledge directly or indirectly influences the pupils for better living.

11. Basis for vocation:

The teaching of biology not only prepares individual for specialization in specific branches, but also some vocation. There are many professions open to them to take up further. They are

- ❖ Scientist
- ❖ Astrologist
- ❖ General Technician
- ❖ Engineer

- ❖ Doctor
- ❖ Teacher
- ❖ Physiologist
- ❖ Physiotherapist
- ❖ Lab technician
- ❖ Electrician
- ❖ Architect
- ❖ Aeronautics
- ❖ Pilots
- ❖ Chemotherapist
- ❖ Space
- ❖ Army Officers

All of them emphasized the teaching of commerce for:

- ❖ The development of process skills like observation, classification, measurement, communication etc.
- ❖ Acquisition and understanding of knowledge, development of problem solving skill, skill of investigation, ability to think logically and to draw conclusions on the basis of experiments.
- ❖ Development of ability to reach generalizations and to apply them for solving everyday problems.
- ❖ Development of understanding of inter-relationships of science and society.
- ❖ To foster creativity in people, enabling them to carry out innovations in commerce.

1.7. NEED AND SIGNIFICANCE OF TEACHING COMMERCE

The science education is aimed for the learner to know the facts and principles of science and its applications, consistent with the stage of cognitive development;

- ❖ Acquire the skills and understand the methods of processes that lead to generation and validation of scientific knowledge
- ❖ Develop a historical and developmental perspective of science and to enable her to view science as a continuing social enterprise;
- ❖ Relate science education to environment (natural environment, artifacts and people), local as well as global and appreciate an issues at the interface of science, technology and society;
- ❖ Acquire the requisite theoretical knowledge and practical technological skills to enter the world of work;

- ❖ Nurture the natural curiosity, aesthetic sense and creativity in science and technology;
- ❖ Imbibe the values of honesty, integrity, cooperation, concern for life and preservation of environment; and
- ❖ Cultivate scientific temper- objectivity, scepticism, critical thinking and freedom from fear and prejudice.

The following discussion about these aims will enable you to understand and plan your teaching-learning in a child centered manner.

1.7.1 Basic Theories for Learning Science - Jean Piaget: Cognitive Theory

Jean Piaget, a Swiss biologist, developed cognitive psychology. Intellectual development in terms of four development stages are:

1. Sensor motor stage (From birth to about 2 years)

Piaget called the first stage of intellectual development as sensorimotor because

- (a) It is characterized by the absence of language, and
- (b) It is limited to direct sensory and motor interaction with the environment.

2. Pre-operational stage (about 2 to 7 years)

While stepping into this age, the child begins to replace direct action in the form of sensory or motor exploration with symbols. This stage is further subdivided into

- (i) The pre-conceptual phase (approximately 2-4yrs)
- (ii) The intuitive phase (approximately 4-7yrs)

3. Concrete operational stage (about 7 to 11 years)

The child reaches a satisfactory level in terms of intellectual development by his thinking becoming quite systematic and logical.

4. Formal operation stage (about 12 to 15 years)

The high order of intellectual functioning developed through this stage, is usually characterized by the presence of the most sophisticated cognitive abilities like the ability to

- a) Build up multiple hypotheses and number of alternate solutions.
- b) Verify all possible solutions in a systematic and logical way.

- c) Generalize and arrive at abstract rules that cover many specific situation.

Application

- 1) The teacher must try to emphasize discovery learning rather than teaching or telling each and every part of information to the pupils.
- 2) Develop symbolic thinking adopted by pupils in their thought process.
- 3) Emphasis on the individualization of education.

1.7.2 Jerome S. Bruner: Discovery Learning

Bruner proposed a theory of instruction quite different from earlier psychologists and the assumptions are:

1. Learning is not simply a stimulus Response(S-R) conditioning but a cognitive process. As a cognitive process it involves the three simultaneous acts:
 - (i) Acquisition of new knowledge or information.
 - (ii) Transformation of knowledge that one already possess and
 - (iii) Checking the pertinence and adequacy of knowledge one has.

2. Curiosity is the basic instinct possessed by every individual.
3. Conceptualization or categorization is the first major step in the process of learning in which Bruner advocated the concept attainment model.
4. The development of cognitive abilities of human begins is a continuous process.

The three development stages are:

- (i) **The enactive stage:** The child gets some aspects of reality by resorting to motor responses.
- (ii) **The iconic stage:** The child is able to think about the objectives or events through mental images.
- (iii) **The symbolic stage:** The child is using symbols in place of images.

5. Knowledge is important.
6. The children should not be made to repeat, reproduce and present the ideas, concepts and Principles but should be encouraged to learn on their own and discover things themselves. This is known as discovery learning.

1.7.3 C. Robert M. Gagne: Events of Instruction

Gagne presented most of the conceptualizations on learning by advancing.

1. **Sign learning:** It is also called classical conditioning where an individual acquires a conditioned response to a given signal. In it the responses are diffuse and emotional and the learning is involuntary.

2. **S-R learning:** It is usually operant conditioning where in the presence or absence of a clue the learner acquires a specific response by a contingency set up either through positive or negative reinforcement situation.

3. **Chaining:** It is skill learning where the person links together two or more units of stimulus-response learning. The chief condition of learning of this type involves the reinstatement of reinstatement of the stimulus response units in a proper sequence.

4. **Verbal association:** It is also a type of chaining but the links are verbal units.

5. **Multiple discrimination:** Here the student learns to distinguish between motor and verbal chains already acquired.

6. **Concept learning:** It requires responding to stimuli in terms of abstract characterization. It involves discrimination between classes and generalization within a class.

7. **Problem solving:** It is the highest type in the hierarchy of learning, where the learner uses principles to achieve some goal. In fact, as Gagne puts it, it involves learning of a higher-order principle, the combined product of two or lower order principles. Thus problem solving implies these internal events which are usually called "thinking".

Application:

1. Teaching task is mainly concerned with helping the students in proper comprehension of the context material.
2. Content material to be organized, arranged and presented in a proper way by following the principles of cognitive learning.
3. Role of teacher as a philosopher, motivator, counsellor, demonstrator and supervisor.
4. Type of educational objectives to be achieved.

1.7.4 Lev Vygotsky: Zone of Proximal Learning

Lev Semyonovich Vygotsky (1896-1934) introduced the social aspect of learning into constructivism. According to him students solve problems beyond their actual developmental level (but within their level of potential development) under adult guidance or in collaboration with more peer.

- ❖ Vygotsky's work includes several key concepts, one of which is the **Zone of Proximal Development (ZPD)** which relates to the gap or difference between what the children can learn unaided and what he or she can learn with the help of an adult or a more capable peer. This idea of assisting the learner is known as **scaffolding**.
 - ❖ When a child works unaided on a task or problem, that individual is said to be at their actual development level. Potential development level is the level of competence a child can reach when he or she is guided and supported by another person.
 - ❖ Another important Vygotskian contribution relates to the development of language as related to thought. This concept is explored in his brilliant book "Thought and Language" in which Vygotsky establishes the explicit and profound connection between oral language (speech) and the development of concepts (mental constructs) and one's conscious awareness of them.
 - ❖ Vygotsky's model of human development has been termed as a **sociocultural approach**. For him, the individual's development is a result of his or her culture. Development, in Vygotsky's theory, applies mainly to mental development, such as thought, language and reasoning processes.
 - ❖ These abilities were understood to develop through social interactions with others and therefore represented the shared knowledge of the culture. These abilities are developed through a process **internalization**.
 - ❖ Internalization describes how children's social activities develop become mental activities. When children participate with parents, teachers and peers, they begin to internalize and process new information.
- 1.7.5 Reflection: A Basic Process 'From Experience'**
- Reflective practice is an evolving concept. In 1930's, John Dewey defined the reflection as a proactive, ongoing examination of beliefs and practices, their origins and their impacts. Till then, reflective practices have been influenced by various philosophical and pedagogical processes. One influence is constructivism, which views learning as an active process where learners reflect upon their current and past knowledge and experiences to generate new ideas and concepts.

Characteristics to cultivate excellence

Reflective teaching is a **strategy for nurturing excellence**. Reflection means thinking. Excellence in teacher education depends to great extent upon the efficiency and professional behaviors of the teacher. So a teacher has to become reflective, which needs a continual process of his professional growth. This professional growth can take place on the foundation of reflective teaching viz. experience, inquiry, transformational leadership of the teacher and curriculum.

According to Dewey, to become a reflective teacher, one must possess the following characteristics for reflective actions, which could nurture excellence in teacher education. They are:

1. **Whole heartedness:** It is willingness, commitment and dedication of the teacher to act according to our beliefs and practices. When the teacher believes in attainment of certain goals he should not hesitate or forget to fight for its attainment.
2. **Broadmindedness:** It should be desire of the teacher to listen to students many sides carefully and to give full attention to alternate possibilities in them. A reflective teacher must keep an open mind about the context, methods and procedures followed in his classroom and constantly inquire and re-evaluate their worth in relation to the student's need and circumstances.
3. **Responsibility:** Teaching involves morals and responsible activities of the teacher. A reflective teacher needs to be aware of his actions, which have definite responsibility selected for the purpose.

1) Definition of Reflection

Bound: Reflection is an intellectual and effective action in which individuals engage to explore their experiences in order to lead to new understandings and applications.

Calderhead and Gates: Reflection is a crucial element in professional growth of teachers.

Chayer: Reflection is looking back and making sense of your practice. Learning from this and using this learning to affect your future action. It is make sense of your professional life.

ii) Strategies for promoting reflection

Kottcamp (1990) has developed a catalogue of tactics to promote reflection among teacher trainees and teacher educators.

The following items have been included in the catalogue.

- 1) **Writing:** Writing to capture teachers' thought process and yield a product that can be evaluated and reformulated.
- 2) **Journal:** Data provided in daily records may be used to search for patterns of means.
- 3) **Case records:** Writing a case record is a structured activity that expands the number of teaching alternatives.
- 4) **Contrived situations:** This includes simulated teaching situations like role plays, case studies, simulations and microteaching.
- 5) **Instrument Feedback:** Scales and inventories can be used to provide teachers with data to clarify the personal meaning teachers attach to their particular teaching action.
- 6) **Electronic Feedback:** Videotaping teachers' actions provides records of events, issues and feelings for reflection on action.
- 7) **Metaphors:** Metaphors can be used to clarify the personal meaning of teachers' attach to their particular teaching action.
- 8) **Platforms:** These are teachers' espoused theories, their stated intentions. Writing a platform engages teachers in a reflective process on the congruity or incongruity between theory and theory-in-use.

Applications and Reflective Practice and Education

- 1) Reflective approach will sharpen teachers' skill in problem solving.
- 2) It helps students need and conducting action research.
- 3) The student teachers should be given opportunity to observe how classroom function achieve on reflection.
- 4) Reflective practice is a strategy for continuous improvement of an action taken to achieve its stated objectives.
- 5) Reflective teaching is used in introducing courses in pre-service teacher education programme to introduce novices to the role of teacher and the task of teaching.
- 6) It is used even more generally in methods course which focus on helping prospective teachers gaining knowledge of theory and then to apply this knowledge under controlled laboratory conditions.
- 7) Reflective practice can be practiced at each step evaluating the learning (both context and methodology), implementing the plan (teaching and interactions) and evaluating the learning (during and after the teaching act is carried out).

1.8 NEED AND IMPORTANCE OF INSTRUCTIONAL OBJECTIVES

1.8.1 Instructional Objectives: Basics

The term objectives is popularly used in all sphere of life, before organizing any activity (for example, organizing the teaching-learning process) we have to have a clear idea about where it will lead us. We should know, in the context of this discussion, what destination (the goal, aim, objective, outcome, end-point or terminal behavior) the student is supposed to reach through the teaching-learning process. We, as teachers or as the organizers of teaching-learning activities, should know the concept, need and role of instructional objectives. At the same time we should be clear in our Kinking about the limitations or weaknesses of instructional objectives. This will help us organize the teaching-learning activities more effectively.

1.8.2 Aims, Objectives and Instructional Objectives

In this sub-section we shall explain various terms used in this course in general and in this unit in particular. These expressions are 'aims', 'Objectives', 'instructional objectives' and 'behavioral objectives'.

Aims:

We use the term 'aims' to mean the broad goals which our educational system (the whole of the Indian society involved) embraces and which you are expected to attain. The aims of education are based on philosophical and socio-psychological aspects of society and culture. The following examples of aims of education will help you understand the concept better.

- ❖ Education to develop patriotism and good citizenship;
- ❖ Education for moral, ethical and spiritual values;
- ❖ Education to fulfil the vocational needs of the individual and society;
- ❖ Education for leisure-time activities such as music, dancing, gardening, reading, hobbies, etc.;
- ❖ Education for improved health knowledge (physical and mental) and practices;
- ❖ Special education for the disabled.

The central purpose which any educational system must be oriented to accomplish concern related to its traditional tasks and

also the new roles that are emphasized by recent changes in the world.

Educational aims are broad and also hold good for the various educational systems (related to school as well as other agencies). The aims in fact express the expectations of the society that should be achieved by the students through the educational system.

We expect from the school as well as other agencies to help students in

- ❖ Building up their personality and character so that they can take their rightful place in the mainstream of the nation;
- ❖ Assimilating vocationally and socially desirable knowledge and skills;
- ❖ Developing ability to think critically and creatively;
- ❖ Developing healthy scientific attitudes towards problems and people; and
- ❖ Motivating them to act according to worthy ideals in the service of community and the nation.

Objectives: Objectives are the milestones to reach the destination i.e. to attain the aim or ultimate goal of education. Objectives are specific, direct and practical in nature, objectives are related to the learning outcomes or change in the behavior of the students. The objectives are meant for both the teacher and the students. Even at the planning stage i.e. before entering the classroom the teacher asks himself, "What changes in behavior of the students can I bring about through this lesson?" Such changes constitute his (teacher's) educational objectives.

An objective is thus

- ❖ a point or an end-view of something towards which action is directed;
- ❖ a planned change sought through an activity;
- ❖ What we set out to do.

Educational objectives, therefore, consist of the changes we wish to produce in the child the changes that take place through education can be represented in:

- ❖ The knowledge children acquire (knowledge);
- ❖ The skills and abilities children attain (skill);
- ❖ The interests children develop (interest); and
- ❖ The attitude children manifest (attitude).

If education imparted to a student is effective, he will 'behave' differently from the way he did before he came to the school. The student would know something about which he was ignorant earlier. He would understand something which he did not understand before. He can solve problems which he could not solve before. He favourably revises his attitudes towards things. Education is thus the process of bringing about changes in the behavior of the students in

- ❖ What they know (Knowledge extends)
- ❖ How they think (Process of thinking changes)
- ❖ How they feel (Prows of feeling changes)
- ❖ How they work (Methods of working change)

1.8.3 Instructional objectives:

There can be two types of objectives as far as the instructional process is concerned. These are

- ❖ Instructional objectives, and
- ❖ Behavioral objectives

The objectives of teachers and students are called instructional objectives. In this context an objective is a statement of a measurable learning that is intended to take place as a result of instruction. Instructional objectives are derived from the terminal behaviours the students are expected to display as a consequence of receiving instruction. Thus instructional objectives are part of the terminal behaviours. But you will find that these terms are quite often used interchangeably. The teacher is accountable for the achievement of instructional objectives.

Instructional objectives are classroom objectives, unique to each course, subject, or a teaching point. These instructional objectives should be so planned that they are consistent with the educational aims. Instructional objectives are set according to the level of the students in a particular class and adopted to the classroom situation. They are formulated in such a way that they are concrete and tangible. The objectives should involve information, skills attitudes and interest% that could be developed through a particular topic or subject taken up in classroom. For example, "to enable the students to develop skill in finding topics in reference books", or "to help him/her know how to dig a compost pit in a village" are quite acceptable instructional objectives.

Instructional objectives thus relate and contribute to the total educational process. Educational aims are indirect, not immediately achievable, while instructional objectives are direct, specific and functional and are directly achievable in classroom within a period or couple of periods. According of Dececco and Crawford (1974) instructional objectives for teachers are narrower in scope than educational objectives. Instructional objectives guide the teacher in the selection, emphasis, and omission of subject matter, materials and activities while preparing lesson plan and classifying instruction.

1.8.4 Behavioral objectives:

In the preceding discussion you have studied that instructional objectives are derived from learning outcomes. Thus instructional objectives can be stated by identifying the product of instruction in terms of observable performance. These outcomes have been referred to as behavioral objectives or terminal performances. Thus when we formulate instructional objectives for students we have to ensure that they are observable or measurable.

An instructional objective certainly tells us about the change(s) we propose to bring about in the student but it will be still clearer if we isolate the critical aspects of a particular change. Statements of objectives in terms of the change in the behavior of the students are called behavioral objectives. Instructional objectives can be transformed into behavioral objectives.

Here is an example of an instructional objective:

"To develop in the student a sense of civic responsibility".

This objective acquires concreteness, clarity and meaning when we get answers to the following questions:

What does a person who has a sense of civic responsibility usually do?

What does a person who has a sense of civic responsibility usually not do?

What kinds of behavior distinguish one who has a sense of civic responsibility from one who lacks a sense of civic responsibility?

An objective, when it is framed in terms of student's behavior becomes tangible and capable of attainment. It is clear from the above illustration that a clearly formulated objective has two dimensions: one deals with the behavior and the other deals with the content area in which the behavior operates.

We have discussed the concept of instructional objectives. The important terms used in this unit have also been discussed.

1.8.5 Why Objectives?

As a matter of fact, the objectives are meant to help us bring about changes in the individual in the desired direction. The achievement of objectives enables the individual to perform certain tasks, develop certain understandings, sustain thinking processes, develop attitudes, add to his stock of knowledge, etc., and thus to lead a happy, productive and socially acceptable life. The question now arises, "How are these objectives responsible for bringing about the required change in the life of an individual?" The answer to this question is presented as follows.

The objectives

- ❖ Provide desired direction to an educational activity;
 - ❖ Distinguish between various aspects of learning;
 - ❖ Focus attention on proper attributes of learning activity;
 - ❖ Determine the nature of an educational activity;
 - ❖ As provide a basis for systematizing or planning an educational programme;
 - ❖ Decide emphasis on educational activity;
 - ❖ Help arrange learning experiences and also evaluation material;
 - ❖ Guide educational decisions - curricular and co-curricular;
 - ❖ Guide & the selection of relevant content;
 - ❖ Give meaning and clarity to the curriculum;
 - ❖ Make learning functional; articulate learning at various levels;
 - ❖ Help discover proper learning situations/context;
 - ❖ Fix priorities in an educational programme;
 - ❖ Help identify weaknesses and strengths in the learning process;
 - ❖ Provide a basis for measurement of growth and development of the child;
 - ❖ Guarantee valid evaluation and curriculum;
 - ❖ Help make educational experiences tangible; and
 - ❖ Define the educational process in totality.
- The above are some of the reasons why we need educational objectives. We can categorise all the above mentioned reasons into three groups. In other words there are three main advantages of instructional objectives. These are:

First, instructional objectives in behavioral terms help plan instruction. The objectives tell us where we are going i.e. what the students will be able to know or do at the end of instruction. Proper statement of objectives will help the teacher plan the steps or procedure to reach the terminal outcomes/behaviours. Thus there is a relationship between instructional objectives and instructional procedures.

Second, the instructional objectives help design performance assessment procedures and help in test construction and curriculum evaluation (refer to Unit 4 Block 1). The assessment of student performance reveals the gap between the expected outcomes and achieved outcomes.

Third, explicit statement of instructional objectives will help the students know what they are supposed to learn or do after an instructional task. The students can direct their energy and time resources accordingly. The students will also know in advance the areas of knowledge. Attitudes or skills on which they will be tested.

We should remember that just writing objectives in behavioral terms does not serve the purpose unless you also know how to achieve them. Some experts or administrators may suggest you various shortcuts (routes) to reach the undertaken objectives but you may end up nowhere. You therefore should know the systematic way for evaluation/assessment of objectives.

1.8.6 Sources of Objectives

Now we will make an attempt to discover the sources for the formulation of objectives. All of us agree that the objectives should be formulated on solid philosophical, psychological, sociological and scientific footings. The utility, appropriateness, practicability and timeliness of the objectives should always be considered. At the same time, we cannot impose any strict limits to the variety of sources. Society, individual and nature of knowledge can be considered to cover most of the areas, if not all. Religion, philosophy or life experiences, for example, may also be used as sources of objectives in their own right. These and many such others can either be considered separately or as part of any of the first three major ones (viz. society, individual and nature of knowledge) mentioned above. All the three can however be accommodated under society. We are of the opinion that the

sources should be viewed from the point of view of practicability and relative usefulness of the objectives.

Let us discuss all the three major sources of formulation of objectives.

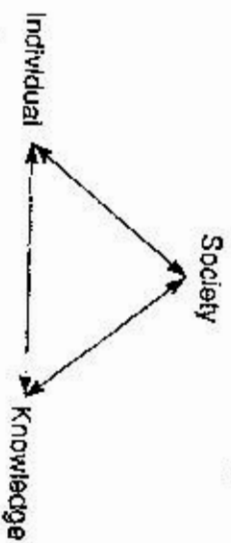
Society: We cannot fail to recognize that objectives of education in the broad sense are determined by society at the local (community) as well as the national level. For instance, there may be such needs as preserving and transmitting cultural heritage, inculcating the democratic values of life and supporting and enhancing the impact of science and technology and other innovations. Social needs are essentially taken into consideration while planning to develop certain competences and qualities in members of society both to develop it (i.e. society) and also to survive in it.

Individual: We have already seen that though society determines the basic requirements of Instructional Objectives education, it is the individual whose needs are reflected through the society. In addition, there are certain specific needs of the individuals. These needs may be grouped as self-development or self-fulfilment. For example, self-development encompasses physical and psychological growth of the students. Psychological growth in turn consists of thinking (cognitive), feeling/ attitudes (affective) and doing psychomotor) aspects. Various theories of learning have tried to explain the process and sequence of growth in the child. Needless to say, psychology of learning is helpful in choosing, grading and sequencing objectives to suit the individual growth. These aspects of human growth will be considered while determining educational objectives. Some educators believe that an individual has some spiritual needs too. It will, therefore, be desirable to provide him some opportunity to fulfil these spiritual needs too.

Nature of knowledge: It is education that differentiates between human beings and animals. One may say that human being are cultured and civilized animals. We know that knowledge is the most important condition to growth of civilization; in the absence of knowledge, there cannot be any growth of civilization. Knowledge has its own categories such as facts, processes, basic ideas, concepts, thought systems, etc. Its classification and organization into systems is known as disciplines or subjects.

Hence we should know that the nature of knowledge (i.e. subject matter) is one of the most important consideration in formulating educational objectives. This is so because different disciplines (i.e. subjects) have different bearing on the learning process. Therefore, in formulating the objectives, the nature of knowledge should be given due importance.

Provides a clear presentation of the three sources of objectives i.e. society, individual and nature of knowledge.



Interrelation between society, individual and knowledge:

Society is not static, and goes on changing and the individual has to adjust in society. But individual has its individuality. The role of knowledge comes to make the individual adjustable in society without sacrificing his individuality. The nature of knowledge is related to what is good for the society and for the individual. Rather we can say that society and individual both are determinants of the nature of knowledge. All the following questions are related to society. The individual and nature of knowledge.

- ❖ May I be a good citizen?
- ❖ May I be a good man?
- ❖ May I be a good scientist?
- ❖ May I be a good artist?
- ❖ May I be a good sportsman?
- ❖ May I be a good soldier?
- ❖ May I be a good politician, etc.

Answers to all the above questions proves the interrelationship between the three.

1.8.7 Criticism of Objectives

You have already seen that there are undisputable gains in using objectives. Education without objectives can be compared to sailing in a rudderless boat. Hence one cannot think of education without objectives. In spite of such an important role of objectives

in the education process, there are, however, some difficulties in using the objectives. A practical difficulty in the use of behavioral objectives relates to analysis and statement of behaviours which is a quite lengthy and difficult process. For complex content, it is difficult to formulate objectives in behavioral terms. For example, you cannot define a creative response in behavioral terms because the particular creative behaviours to be developed cannot be easily identified. The statement of objectives in behavioral terms is further criticized because of the following limitations:

First, we know that there are varied sources for developing the objectives. These sources are not static. Since society is changing day by day, its needs and expectations are also changing. An individual's interests, needs, age level, mental age level and the subject matter, its nature and its relevance there are a number of factors responsible for developing objectives e.g. calculative ability is now completely replaced by very simple calculators.

Second, the method of formulation of objectives also differs from individual to individual. There is no single procedure for the formulation of objectives used by all the class, e.g. interests of individual, age and maturity.

Third, even though we consider the change in behavior of an individual the end-product of these objectives, these end-points are different for different students, not only in amount but also in type.

Fourth, quite often the objectives are vague and overlapping while it is claimed that these are hierarchical in nature. There is no definite cut off point where an objective may end and the next objective begins, e.g. knowledge and understanding, no cutting point between the two.

Fifth, it is very difficult to frame curricula in accordance with the objectives. Because, for example, it is very difficult to give limits and boundaries to any trait such as Truth, Goodness, Beauty, Honesty, Intelligence, etc. While the objective is to achieve these traits, it is a very difficult task.

Sixth, the evaluation part i.e. to frame the objective-based test items is also a difficult task. It is an act that test items based on knowledge (recall or recognition) are very easy to frame but items based on evaluation, originality of thoughts and ability to imagine or ability for abstract thinking are difficult to frame.

For example:

- ❖ What is an atom? Easy to frame

What is fission or fusion? Difficult to frame

❖ What is Archimedes's principle? Easy to frame

Why does a sports man take a dive into water head first and not feet first? Difficult to frame.

With sufficient practice, experience and familiarity with the process of curriculum construction, one can overcome these difficulties to a considerable extent. The experts opinion can be a way that casts the final judgment regarding the relevance in the objectives, curriculum and testing. Though it is very difficult to obtain entirely valid data related to human behavior, it will nevertheless be a great shortcoming not to use the objectives in preparing curriculum, designing instruction and testing materials and students.

1.9 BLOOM'S TAXONOMY OF EDUCATIONAL OBJECTIVES: AT COGNITIVE, AFFECTIVE AND PSYCHOMOTOR LEVELS

Evaluation is a process of ascertaining the nature and quantum of change and should necessarily be based on, or be directed towards such a goal. An educational objective is often limited to what is mentioned in the prescribed syllabus or to what the teacher does in the classroom.

1.9.1 Educational Objectives

An educational objective may better be defined "as a desired change in behavior in a person that we try to bring about through education" - E.J.Frust

These changes have the basic characteristic of direction and nature (quality and extent). In other words, it may mean

- ❖ A dimension of learning.
- ❖ Worthwhileness of a pattern of learning for realizing it, and
- ❖ Level of learning to be attempted.

1.9.2 Instructional Objectives

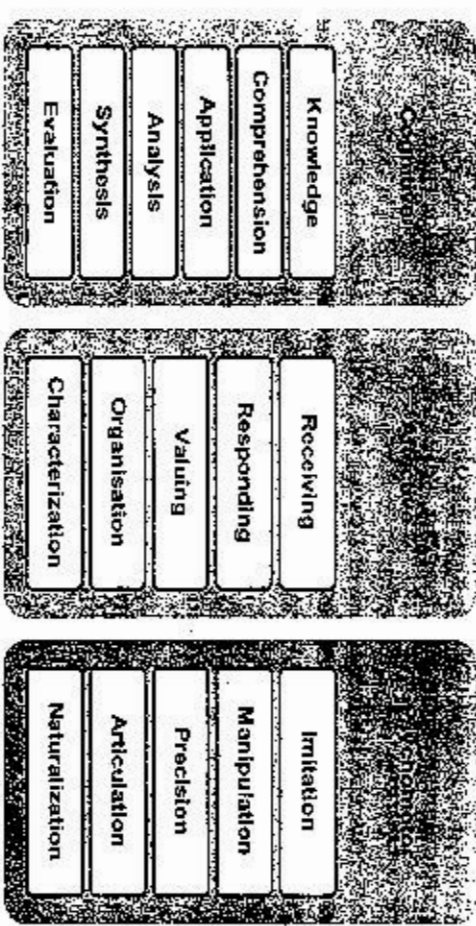
Every day, teachers make wide variety of instructional decisions that directly affect their students' learning. These decisions range from the choice of materials pacing and sequencing of activities, to ways of reinforcing pupil's learning and means for assessing whatever the students have learnt. Different types of learning require different learning experiences and hence different types of objectives.

Teaching learning situation at any level involves three major components namely the teacher, the learner and the subject matter to be processed. The casual observation of classroom at the school level would reveal the typical picture of a teacher with a group of students and the teacher enabling them to process knowledge and concepts.

Instructional objectives are specific and are behavioral in nature. These are mainly based on specific observable or measurable goals in pupil's learning.

1.9.3 Bloom's Taxonomy of Educational Objectives

Bloom and his associates developed a taxonomic code of educational objectives. They adopted a division corresponding to three primary aspects of pupil's growth viz. cognitive, affective and psychomotor. Within each domain they arranged the objectives in logical and psychological order i.e. the sequence of the development of mental abilities have been maintained.



1.9.4 Cognitive Domain

The cognitive domain represents the intellectual component of mental life. It is the most important aspect with respect to education. You will appreciate the sequence in which these mental abilities are classified under this domain. These are as follows:

❖ Knowledge

Memory or remembering is the lowest level of intellectual structure model. Knowledge involves the recall or recognition of specifics, universal, pattern, structure etc. for example, recalling the names of animals, recognizing them as herbivorous or

carnivorous, recalling their food habit, type of food they eat and type of teeth or alimentary canal they have.

❖ **Comprehension**

Comprehension is considered at a higher level than knowledge but is also the lowest level of understanding. An individual is expected to go beyond the level of recall and recognition. Pupils should be able to understand the meaning of the concepts, because facts and concepts describe process. For example, relating the type of food with the structure of teeth or alimentary canal of herbivorous or carnivorous animals or describing the process of 'raining' by relating the concept of 'evaporation' and 'condensation'.

❖ **Application**

While applying, an individual is expected to use his knowledge and understanding of the facts, concepts, principles, ideas, or procedure in a particular situation or for solving any particular/specific problem e.g. selecting the food for their pets.

❖ **Analysis**

It is the higher level of cognitive ability. Here an individual is required to analyze a meaningful statement into its various components. Pupils should be able to identify the concepts and their inter-relationship in the statement. Analysis will lead to establish the relationship between various statements in a meaningful communication.

❖ **Synthesis**

It is a complex ability at a higher level of cognition. It involves the ability of an individual to put together the elements or parts in a way that new pattern emerges out. It is an ability to give new shape or structure to statements or procedure. It has an inherent element of creativity.

❖ **Evaluation**

The highest level of cognitive structure is the ability to evaluate or judge. It is also the most complex ability which involves all the other abilities. It enables a person to judge a material, product or process against a standard and to establish the worth of it.

All these abilities in the cognitive domain are in a logical order. Application of knowledge is not possible until an individual has a sound understanding about the material and the process to be

used. Also, for developing this understanding, knowledge is the prerequisite.

1.9.5 Affective Domain

There is another important aspect of human nature which includes the interests, likes and dislikes, feelings, behavior and attitude etc. psychologists have tried to identify and arrange three qualities in a gradual manner like cognitive abilities. These qualities are classified under the category of Affective Domain. These are the behavioral abilities which lead towards the formation of attitude, the complex and ultimate behavior norm. These are as follows.

❖ **Receiving**

The lowest level under the affective domain towards the development of attitude is Receiving, i.e. the ability of an individual to receive information. Attention of an individual is an indication that he/she is ready to receive information. Awareness of the sources of information and willingness to receive the information are the sub-levels of this category.

Eg. Awareness of Science Centre/ Information Centre Library/Video lib. etc. in the neighborhood.

❖ **Responding**

Regularity in attention and motivation leads to Responding. This ability is represented by interests. Interest is a tendency to respond to a particular object/situation.

Eg. Visits resource centers collecting/reading materials/obtaining membership of centers.

❖ **Valuing**

The next level towards behavior formation is achieved when an individual is able to internalize the sense of worth and identification with it. Individuals set guidelines for controlling their own behavior. Acceptance of a value, preference for a value and finally the commitment to a value are steps towards character building. Take decision about the activities of the groups and analyzing and assessing the issue.

❖ **Organization**

An individual's behavior or attitude is not ordinarily motivated by a single or isolated value. A system of values form the complex behavior, represented by general behavior, a code of conduct. There should be an organization of values for balanced conduct.

Decision about conduct in real life situations forming opinions about major issues in reality.

❖ *Characterization*

It is the highest level of the character building or attitude formation of an individual. The values are imbibed and the individual has a consistent philosophy of life.

Set attitude basis for → Character building → leads to Personality development

Whatever function an individual performs is a result of neuromuscular coordination. As the level of coordination goes up, the action becomes more rapid, speedy and automatic. The action abilities are classified under

1.9.6 Psychomotor Domain

Five categories have been identified under this domain.

❖ *Imitation*

It is the lowest level of psycho-muscular activity. It starts an impulse and may grow into an overt act with the capacity to repeat the performance. Though this ability is found in early childhood years of individuals it is also carried over, to some extent, in the later years. This is also the first essential skill for training the students for practical and experimental exercises. They learn the skills first by observing others (teachers) performing them.

❖ *Manipulation*

It is the next higher level of motor coordination. It involves selecting a certain action in preference to others, following directions and acting accordingly. Selection is important in manipulation.

❖ *Precision*

Practice or repetition of performance will result in decreasing the faults in performance. Precision is related with speed and refinement giving the learner the ability to control his/her action in response to the requirement.

❖ *Articulation*

On attaining the ability of articulation, individuals will be able to handle many actions in unison. This ability involves the coordination in action.

❖ *Naturalization*

Perfection in performance or in any action is the final level in psychomotor domain. On attaining perfection an individual

requires minimum time for completing a perfect task his actions become mechanical and without any conscious thinking or planning.

For teaching/learning science, these abilities are very essential and need to be developed from primary stage. One can learn science only by doing it. Experimentation, and measurement are the basic skills for learning science. Precision is required in measurement and observation, articulation and naturalization are required in planning, designing and performing experiments.

1.10 REVISED BLOOM'S TAXONOMY

In early 1940's, the cognitive researchers and psychologists had stated, based on cognitive and human information processing theories, that a human's mind applies different mental processes while he or she manipulates, develops, insights, stores, and retrieves information (Gagne, 1977; Guilford, 1956; Lindsay & Norman, 1977; Piaget, 1952; Rothkopf, 1966; Rumelhart, & Ortony, 1977; Wittrock, 1974). These mental processes differ in terms of types, such as memorization, comprehension, discrimination, analysis and the like, and also in terms of level of difficulty, such as simple, medium or complex. Accordingly, a number of instructional psychologists have utilized the results of cognitive research and applied its principles into field of education (Bloom, 1956; Guilford, 1967; Gagne, & Briggs, 1979; Merrill, 1983). They aim to help teachers and curriculum developers to design instruction that will induce students to use different forms of mental processes, on different levels of difficulty, so that the whole instructional process will be remarkably enhanced.

One of the most prominent instructional psychologists who tried to apply the cognitive principles was Benjamin Bloom. Bloom (1956) published a taxonomy of educational objectives within the cognitive domain. Bloom and his associates classified different forms and levels of learning based on mental processes that students involved in while they learn. Bloom's taxonomy gained immediate popularity; however, after almost six decades of using Bloom's original taxonomy, some educators began to wonder whether the taxonomy was still valid to this age that characterized of lot of research and studies on intellectual skills and human thinking and learning. One of those educators was Lorin Anderson, a former student of Benjamin Bloom. Anderson and Krathwohl (2001) revised Bloom's taxonomy to be more adaptive to our current age by proposing another taxonomy that

will meet curriculum designers, teachers, and students needs better than the Bloom's one. However, based on a thorough assessment we contend that Anderson and Krathwohl's (2001) revised Bloom's taxonomy still fails to match the cognitive psychologists recently discovered types and difficulty levels of mental processing (Bloom, 1956; Darwazeh, 1995, 2011; Lindsay & Norman, 1977; Merrill, 1983; Tuckman, 1992; West, Farmer, and Wolff, 1991). Therefore, based on a current review of the literature on cognition, and particularly based on Merrill's Component Display Theory (1983), further revision to the revised Bloom's taxonomy is appropriate.

1.10.1 The original Bloom's educational taxonomy

Bloom et al (1956) published the taxonomy of educational objectives: a cognitive domain. Bloom et al (1956) classified *forms and levels of learning* based on cognitive processes that learners involved in when they learn. Bloom considered his initial effort to be a starting point, as evidenced in memorandum from 1971 in which he stated "Ideally each major field should have its own taxonomy in its own language – more detailed, closer to the special language and thinking of its experts, reflecting its own appropriate sub-division and levels of education, with possible new categories, combinations of categories and omitting categories as appropriate".

1.10.2 Anderson's revised taxonomy as a match to Bloom's taxonomy

Anderson (1990), a former student of Bloom, updated and revised the taxonomy reflecting relevance to 21st century work for both students and teachers as she said (Anderson & Krathwohl, 2001). Anderson changed the taxonomy in three broad categories: terminology, structure and emphasis (Forehands, 2005). Anderson modified the original terminology by changing Bloom's categories from nouns to verbs. Anderson renamed the knowledge categories into remember, comprehension into understanding and synthesis into create categories. Anderson also changed the order of synthesis and placed it at the top of the triangle under the name of *Create* (Taylor & Francis, 2002). Thus, Anderson and Krathwohl's (2001) revised Bloom's taxonomy became: Remember, Understand, Apply, Analyze, Evaluate and Create (Figure 1).

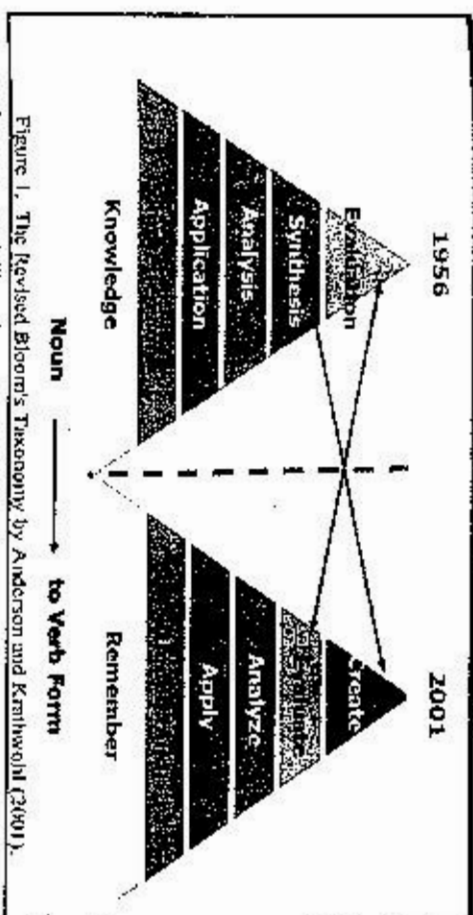


Figure 1. The Revised Bloom's Taxonomy by Anderson and Krathwohl (2001).

Anderson and Krathwohl (2001) also made structural changes to the original Bloom's taxonomy. Anderson considered two dimensions in the revised taxonomy instead of one, a products dimension. The two dimensions are:

1) *Knowledge* (or the kind of knowledge to be learned) and 2) *cognitive process* (or the cognitive processes to be used in acquiring knowledge). The intersection of the knowledge and cognitive categories form 24 separate cells as represented in Figure 2. Based on Anderson's perspective, the Knowledge Dimension on the left side is composed of four kinds: Factual, Conceptual, Procedural, and Meta-Cognitive knowledge. The Cognitive Process Dimension across the top of the grid consists of six levels: Remember, Understand, Apply, Analyze, Evaluate, and Create

	The Cognitive Process Dimension					
	Remember	Understand	Apply	Analyze	Evaluate	Create
Factual Knowledge	List	Summarize	Classify	Order	Rank	Compile
Conceptual Knowledge	Describe	Interpret	Experiment	Explain	Assess	Plan
Procedural Knowledge	Tabulate	Predict	Calculate	Differentiate	Conclude	Compose
Meta-Cognitive Knowledge	Appropriate use	Execute	Construct	Achieve	Action	Actualize

Figure 2. Knowledge and Cognitive Dimensions of Bloom's Taxonomy as revised by Anderson.

1.10.3 Notes, Comments And Justifications on the Revised Bloom's Taxonomy

Based on an analysis of the Anderson and Krathwohl (2001) revision of Bloom's original 1956 taxonomy, the authors recommend further revisions based six reasons:

First, Anderson's revised Bloom's Taxonomy omits several points within the cognitive dimension and the knowledge dimension.

Second, The organizing process is missing.

Third, The sequence of cognitive processes should be reconsidered.

Fourth, Remembering level should be sub-divided.

Fifth, The difficulty level for analysis, organize, and application should be replaced.

Sixth, A principle type of knowledge has been omitted.

First: The revised Bloom's Taxonomy, which had been proposed by Anderson and her associates, omits several points either in the cognitive process dimension, or in the knowledge dimension. The review of the literature on Meta-Cognitive processes which have been conducted after the appearance of Bloom's taxonomy in 1956 (Baker & Brown, 1984; Bondy, 1984; Brown, 1981; Burley, Brown, & Saunders, 1985; Darwazeh, 2004; Derrnody, 1988; Flavell, 1976; Gagne, 1977; Gilbert, 1986; Haller, et al. 1988; Kaufman, & Randlett, 1983; Lindsay & Norman, 1977; Paris & Winograd, 1990; Pressley, Borkowski, & Sullivan, 1985; Scar, & Zanden, 1984; Tregaskes, 1989; Wham, 1987; Wade, & Reynolds, 1989) and studies on instructional design (Darwazeh, 1994; 2001, 2011; Gagne, Briggs, & Wager, 1992; Merrill, 1983; Reigeluth, 1983; West, et al. 1991), stated that the Meta-cognitive process is a process of thinking rather than a type of knowledge, with the primary definition of meta-cognition is thinking about thinking.

Second: Another note on Anderson's Revised Bloom's Taxonomy is related to the levels of cognitive processes. The cognitive dimension is missing a major process that the mind performs from the earliest stages of human development. It is the *Organizing*

process, which was named by Bloom as a synthesis process. Therefore, we cannot integrate this process under the application process and put the two of them in one category. We cannot also look at the organizing process as a creative process like Anderson did in her revision. The organizing or synthesizing process should be separated in a different category as Bloom did in his original taxonomy.

Third: The sequence of cognitive processes should be reconsidered. Anderson and Krathwohl (2001) looked at the organizing or synthesis process as a creative process, which we dispute, because the synthesis process is easier to perform than the creation process (Figure 1). Creation needs form the person to give something new and original, whereas, the synthesis needs to put the elements or parts together according to a certain model or principle. In this case, the synthesis is a prerequisite of the creation process. Based on the perspective of synthesis as a prerequisite for creation, we differ from Anderson, et al. in their proposed taxonomy that considers the synthesizer process as an organizing process. The organizer usually comes after the analysis process.

Fourth: The *Remembering* level should be sub-divided. Anderson and Krathwohl's (2001) Revised Bloom's Taxonomy did not differentiate between two levels of remembrance, (i) remember specific information and (ii) remember general information. Merrill (1983) has clearly differentiated in his Component Display Theory between the two levels based on the content type: a) remember specific information and b) remember general information. Merrill defined the specific information as a kind of knowledge or data which cannot be generalized or transferred to another situation, because, it is just valid on itself such as, dates or events, names, symbols, terms, labels, examples, lists, instances; whereas, the general information is a kind of knowledge that can be generalized and applied to more than one situation, such as concepts, principles and procedures. Merrill also claimed that remembering specific information is easier to recall or recognize than remember general information, and so it should actually come before remembering general information.

Fifth: The difficulty level for analysis, organization, and application should be replaced. The application process is more

difficult to perform than the analysis or organizing processes when encountering a situation for the first time, so the application should come afterward. The reason is, the application process defined by both Bloom (1956) and Merrill (1983) as the usage of previous acquired knowledge (generalities) in new situations. The contention here is that learners can only use or apply the acquired knowledge in a new, novel, or strange situation unless he or she inspects, analyzes, organizes, and re-organize its components. These processes of analyzing and organizing will help him or her to see where the previous learned knowledge will fit in a new situation, thus, to use and apply them correctly.

Sixth: A principle type of knowledge has been omitted. The Revised Bloom's Taxonomy is missing a major type of knowledge, the principle type or relationships knowledge (West, et al. 1991). Anderson (Anderson & Krathwohl, 2001, p. 44) put the principle under the conceptual type of knowledge, which is not true according to Merrill's Component Display Theory (Merrill, 1983). West (West, et al. 1991) and Gagne (Gagne, 1977, Merrill (1983, pp.287-288) in his theory had identified four types of knowledge or content: facts, concepts, principles, and procedures. Therefore, the principle is one type of the content. The contention here is that Bloom's Revised Taxonomy (Anderson, 1990) should include the principle or relationship type of knowledge because it is different than the facts, concepts, and procedures. The revised Bloom's Taxonomy should also move Meta-Cognitive knowledge to the cognitive dimension, because the meta-cognitive is a kind of mental process rather than a kind of knowledge.

1.10.4 Suggested Revision to the Revised Bloom Taxonomy

We propose the following taxonomy based on the original 1956 Bloom's Taxonomy and the Anderson and Krathwohl (2001) Revised Bloom's Taxonomy. Our new proposed taxonomy has two dimensions: *Dimension One: Cognitive Processes*, which range from simple to complex, according to the difficulty level of the mental process vertically, and ranging from simple to complex horizontally according to the number of items in which a mental process is involved. *Dimension Two* according to Merrill's theory of four types of knowledge: facts, concepts,

principles, and procedures. The content contained within the cells of Figure 3 represents the behavior that the learner exhibits during knowledge construction after learning has occurred.

Accordingly, the content contained within the cells of Figure 3 represents the cognitive process (behavior) that the learner can exhibit during knowledge construction after learning has occurred. Check marks represent that students can exhibit this kind of behavior related to this certain type of knowledge, and X marks represent that students can't exhibit. The rationale is that once the facts are a kind of specific information knowledge which can't be generalized to more than one new situations, thus, the student can't comprehend, apply, evaluate, or create them. A learner can remember them on specific level of learning only, but not on a general level. The student also can discover the facts but can't create them, because they are existed in the world already. The student can also recognize and analyze the elements of facts either names, symbols, dates, labels, and organize and re-organize them based on a certain principle. For example, the student can re-order the list of names alphabetically, once recognized and analyzed. Finally, a student can accept the fact either he likes it or dislikes it. So he can evaluate facts, which can be accepted or rejected (Figure 3).

	The Cognitive Process Dimension							
	Facts' Remembrance	Generalities' Remembrance	Comprehension	Analyzing	Organizing	Application	Evaluation	Creation
Factual Knowledge	☐	x	x	☐	☐	x	x	x
Conceptual Knowledge	x	☐	☐	☐	☐	☐	☐	☐
Principles Knowledge	x	☐	☐	☐	☐	☐	☐	☐
Procedural Knowledge	x	☐	☐	☐	☐	☐	☐	☐

Figure 3. Knowledge and Cognitive Dimension of Bloom's Taxonomy Revised by Anderson

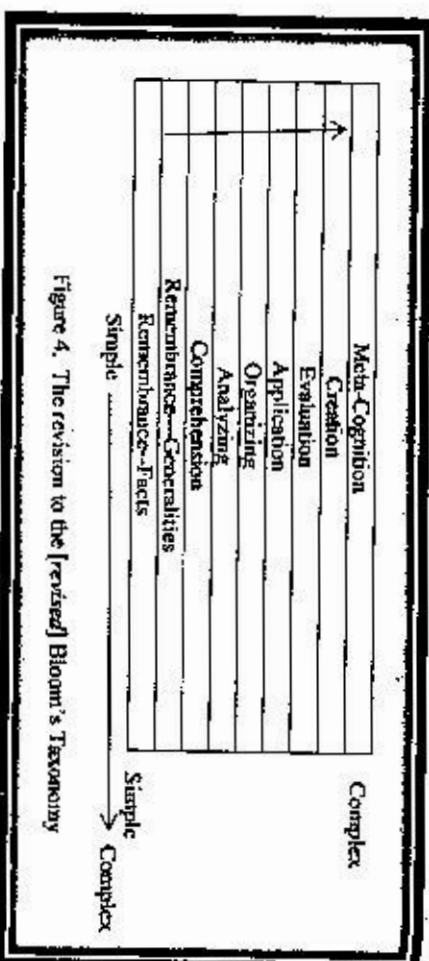


Figure 4. The revision to the [revised] Bloom's Taxonomy

The purpose for this proposed revision to Anderson's (2001) revised taxonomy is to be used as in a broader and more comprehensive view by educators, teachers, instructors, professional trainers, and curriculum planners. Further, evaluators who intend to promote students' thinking and skills in order to make them good thinkers, creators, discoverers, builders, and critical thinkers, will be in a better position to serve their societies and nations in a good and productive manner in order to live within the Information Age.

1.11 CORRELATION OF COMMERCE AND ACCOUNTANCY WITH OTHER SUBJECTS

The close linkage and correlation of commerce with a number of subjects have been highlighted here. It has been pointed out there that the scope of Commerce is very vast and it includes all aspects of modern life. It involves a study of many related disciplines apart from its own specialized branches. At the same time the correlation of commerce with three subjects namely Economics, Geography and Mathematics are specially significant. Hence these are discussed below

Correlation of Commerce and Accountancy with reference to Economics is divided into four parts namely, Production, Exchange, Distribution, and Consumption. Of this sub-division of economics, commerce is connecting with the whole of exchange and a part of production. Commerce includes all those activities that take place in the flow of goods between producers and

consumers. In Economics this aspect is known as "exchange the exchange parts of Economics is exclusively the contribution of Commerce.

Commerce provide us with the theoretical basis and practical suggestions helpful to facilitate the exchange of goods and services. These ideas are profusely used by economists to make exchange a scientific endeavor.

The goods produced should reach the hands of consumers to satisfy their wants for this their possession and ownership must be transferred from the producers to the consumers. The commercial activities involved in this transfer include the activities of various middlemen involved in the exchange. These commercial activities complete the full cycle of economic activities. In short, we can say that Economics is the mother of commerce. As such studies in commerce have to make use of ideas from economics. In turn, the modern development in commerce helps to enrich the theory and practice of economics, thus they supplement and complement each other.

1.11.1 Correlation of Commerce and Accountancy with reference to Mathematics

Commerce means all the activities that take place in the exchange of goods from the producer to the consumer. A business concern keenly observes these activities and records them in the books of accounts. A skilful accountant prepares accounts in such a way as to show up all the deficiencies and strong points of the firm concerned. An accountant applies the fundamental arithmetical processes in preparing the accounts. Further in the field of sale tax, income tax, etc. Knowledge of mathematics is essential. Statistical techniques are profusely used for interpretation of data and for making predictions. All these make the relation between commerce and mathematics evident.

Correlation of Commerce and Accountancy with reference to Geography studies about the earth. And it's climate, land scope, etc. The producers are producing agricultural goods and mineral products in tune with the climatic conditions of the region they inhabit. It is not possible to produce all the commodities necessary for human living at the same place. The raw materials required for any commercially significant commodity have to be collected from various places and made available in the center of production. The availability of such materials is always based

upon the geographical conditions. For example, a flood or a drought can fail even the functioning of a commercial endeavor. From these aspects it is clear that, there is very close relationship between commerce and geography.

1.11.2 Correlation of Commerce with Business Management

The word management when used as a collective noun, refers to all those who manage activities within an organization. Management brings together all the factors of productions related to the endeavor and acts as a coordinating work force. In the words of John F. Mee, "Management may be defined as the art of securing maximum results within a minimum of effort, so as to secure maximum prosperity and happiness for both employer and employee and give the public the best possible service". Industry produces goods, which are meant of consumers. This means that consumer must beneficially enjoy the fruits of production. All activities ensuring the free flow of goods from the producer to the consumer are considered as elements of commerce. These include transportation, insurance, and advertising, banking, warehousing, etc. Each of these elements demands effective management. The future of any business that involves production and sale of goods and services depends on efficient management. So it may be said that there is an inseparable relation between commerce and management.

1.12 VALUES OF TEACHING COMMERCE AND ACCOUNTANCY

Aims are considered as conscious purpose and goals. Values are the outcomes or results achieved after teaching according to these aims. Aims are ideals which are not based on experiments. Values on the other hand spring as a result of experimentation or after putting aims into practices. Values are based on reality and the aims on philosophical consideration. Values of the study of commerce may be divided into the following categories

1.12.1 Practical or Utilitarian value

Commerce education has great significance in preparing individuals and the society in general for taking up and indulging in commercial activities efficiently. This is the period of globally relevant industrial activities and subsequent commercial activities involving various aspects of trade. The economic stability of any nation depends upon how scientifically and systematically these are organized. Commerce education provides us with the

understandings, insights, attitudes and skills for taking up such activities, also, it gives proper guidance for systematically extending customer services, thus meeting the everyday requirements of peoples in terms of various commodities essentially required for the very sustenance of life. This may be considered as the practical value of commerce education.

1.12.2 Social value

The study of Commerce should enable an individual to appreciate that man is a social being and he must play an important role in bringing about social progress. Bad business dealings corrupt methods, smuggling and tax evasions are examples of anti-social values which should be avoided as outcomes of a study of commerce. Profit motive should not be the end and be all of all business enterprises. Comfortable life in a society depends upon how efficiently and fairly the dealings associated with commercial transactions like trade, distribution, services, etc. related to goods take place in the society. Performing all these dealings with a social commitment depends upon how well the members of the society get educated in these. Commerce education meets this social value also.

1.12.3 Cultural value

Culture is evidenced by correct socially desirable ways of thinking and doing. It is closely related with value-orientations. Commerce is an important aspect of the life of a nation. It explains the nature of the society in its special aspect of business and material prosperity. Commerce attempts to relate production with consumption. Fair dealings, good salesmanship, honesty in business and pleasing manners all go to make up the good cultural traits of the individual in the commercial field. Several philanthropists in India from the business community have set up a large number of cultural and educational institutions in the country.

1.12.4 Moral value

Commerce is a valuable discipline for each country's development. It can help in solving various economic problems as unemployment, overproduction, and unequal distribution of wealth, ineffective economic policies, low living standard, poverty and many other social evils. There is no aspect of human life, where the knowledge of commerce is not useful. Whether there may be state men, householders, businessmen, producers,

labourers and manufacturers, they can't make progress without the knowledge of commerce education. Thus looked at from every point of view, it is an extremely useful and important subject.

1.12.5 Disciplinary value

The word 'discipline' has a special connotation in commerce. It is not 'interim of order' and 'authority'. It is to be interpreted in terms of intellectual traits only a study of Commerce helps to develop the powers of knowing understanding and application. It also develops certain skill how to write good business letter etc. modern commercial theory and practice applies scientific methods for analyzing various components, planning various tasks, gathering a variety of data, systematically processing and interpreting them, etc. Commerce education reflects all these characteristics of scientific thinking and hence helps the learners of the science to acquire systematic ways of thinking and doing. This is the disciplinary value of commerce education.

1.12.6 Vocational value

The teaching of book keeping and accounting may satisfy the vocational aspiration of students because of the skill development nature. An understanding of book keeping principles together with accompanied skill may help students to become useful workers in business houses and industrial office. The educational objective of teaching commerce and accountancy are mainly determined by the competencies. The Commerce Programme is to prepare students for immediate employment and for higher education. But it is significant that the vocational aspect of commerce education should not be restricted to income. It should also inculcate some other qualities such as job satisfaction, social responsibility, social ethics, etc. Moreover the commerce education should enhance the vocational competency of an individual also.

1.13 RECENT DEVELOPMENTS IN COMMERCE

Banking, marketing, insurance, communication, trade, etc. can be considered as the major areas of commerce. These are the fast developing areas as far as commerce is concerned. Now let us see the recent development that takes place in these domains of commercial activity.

Banking: With the advent of computers, banks can now offer to their customers a variety of new services such as

- (i) Electronic Funds Transfer System (EFTS)
- (ii) Automated Teller Machines (ATMs)

- (iii) Debit Cards
- (iv) Credit Cards, and

(v) Core Banking. Other most modern facilities offered by banks are Tele Banking, and Internet Banking. In the former, banking is carried on over telephone. In the later, Internet facility is made use of.

Marketing: In this age of information explosion, the computer and communication technology play a vital role in marketing. E-commerce is the practice of buying and selling products and services over the internet, utilizing web, electronic data interchange, electronic fund transfers and smart cards. E-commerce enables us the online marketing and network marketing.

Insurance: Insurance is a method of averaging risks. This is an agreement between the insurer and the insured. Entry of private companies is noted as the major advancement in the field of insurance.

Communication: The fruits of fast changing technology are enjoyed by communication. This is evidenced by the successful application of a number of efficient devices of communication. Spectacular developments in this area are fax, internet, e-mail, extranet, video conferencing and teleconferencing.

Trade: Trade involves buying and selling of goods and services. Trade can be of two types- internal and external. Internal trade indicates transfer or exchange of goods and services within a country and external trade indicates trade among nations. Online trading is a recent development in the field of trade.

Business activities are fast changing in this dynamic world. During the 1990's a concept has evolved in the area of commerce called Business Process Outsourcing (BPO). BPO is a system of getting a business task done through an outside agency.
